Python String Interview Questions

25 Mixed Practice & Coding Assessment Problems

Topic: String Manipulation **Total Questions:** 25 **Levels:** Basic → Advanced

📌 Level Distribution Breakdown

Level	Questions	Focus Area
Basic	1 – 8	Character counting, basic string traversal & properties
Intermediate	9 – 16	Word manipulation, duplicate removal & frequency analysis
TCS / Service-Based	17 – 21	Anagrams, word reversal, rotation & pattern logic
Advanced / Challenge	22 – 25	String compression, password validation & regex/parsing logic

1. Count Characters

🟢 Basic

Write a Python program to find the total number of characters in a given string.

> **Input:** hello world | **Output:** 11

2. Reverse a String

🟢 Basic

Write a program to reverse a given string without using a built-in reverse function.

> **Input:** python | **Output:** nohtyp

3. Palindrome String

🟢 Basic

Write a program to check whether a given string is a palindrome.

> **Input:** madam | **Output:** Palindrome

4. Count Vowels

🟢 Basic

Write a program to count the number of vowels (a, e, i, o, u) in a string.

> **Input:** interview | **Output:** 4

5. Count Consonants

🟢 Basic

Write a program to count the number of consonants in a given string.

> **Input:** coding | **Output:** 4

6. Count Digits

🟢 Basic

Given a string containing letters, digits, and special characters, count the number of digits.

> **Input:** py123th4 | **Output:** 4

7. Count Spaces

🟢 Basic

Write a program to count the number of spaces in a sentence.

> **Input:** Python is fun | **Output:** 2

8. Character Occurrence

🟢 Basic

Given a string and a character, find how many times that character occurs in the string.

> **Input String:** banana, **Target Character:** a | **Output:** 3

9. Remove Spaces

🟡 Intermediate

Write a program to remove all spaces from a given string.

> **Input:** p y t h o n | **Output:** python

10. Swap Case

🟡 Intermediate

Convert uppercase characters to lowercase and lowercase characters to uppercase in a given string.

> **Input:** PyThOn | **Output:** pYtHoN

11. Count Words

🟡 Intermediate

Given a sentence, find the total number of words.

> **Input:** Data Structures and Algorithms | **Output:** 4

12. Longest Word

🟡 Intermediate

Given a sentence, find and display the longest word.

> **Input:** Machine learning is powerful | **Output:** powerful

13. Remove Duplicate Characters

🟡 Intermediate

Remove duplicate characters from a string while preserving the order of their first occurrence.

> **Input:** programming | **Output:** progamin

14. Character Frequency

🟡 Intermediate

Write a program to find the frequency of every character in a string.

> **Input:** code | **Output:** c: 1, o: 1, d: 1, e: 1

15. First Repeated Character

🟡 Intermediate

Find the first character that appears more than once in a given string.

> **Input:** programming | **Output:** r

16. First Non-Repeating Character

🟡 Intermediate

Find the first character that occurs only once in the string.

> **Input:** swiss | **Output:** w

17. Anagram Check

🔵 TCS Level

Given two strings, check whether they are anagrams of each other.

> **Input:** listen, silent | **Output:** Anagram

18. Reverse Words

🔵 TCS Level

Reverse the order of words in a sentence.

> **Input:** Python is easy | **Output:** easy is Python

19. Reverse Each Word

🔵 TCS Level

Reverse every word in a sentence while keeping the original word order unchanged.

> **Input:** Python is easy | **Output:** nohtyP si ysae

20. Remove Vowels

🔵 TCS Level

Remove all vowels from a given string.

> **Input:** education | **Output:** dctn

21. String Rotation

🔵 TCS Level

Given two strings, check whether the second string is a rotation of the first string.

> **Input:** ABCD, CDAB | **Output:** Rotation

22. Most Frequent Character

🔴 Advanced

Find the character that occurs the maximum number of times in a string.

> **Input:** success | **Output:** s

23. String Compression

🔴 Advanced

Compress consecutive repeated characters in a string.

> **Input:** aaabbc | **Output:** a3b2c1

24. Password Validation

🔴 Advanced

Write a program to validate a password based on the following criteria:

> - Minimum 8 characters in length
> - At least one uppercase letter
> - At least one lowercase letter
> - At least one digit
>
> **Example 1:** Pass1234 → Valid Password
> **Example 2:** pass1234 → Invalid Password

25. Real-Time Email Validation

🔴 Advanced

A registration system accepts an email address only when it satisfies all of these conditions:

- Contains @
- Contains .
- @ appears before .
- There is non-empty text before @

Example 1: user@gmail.com → Valid Email
Example 2: @gmail.com → Invalid Email